Morgan Stanley | RESEARCH

IDEA

May 26, 2026 01:40 AM GMT

Biopharma | North America

Lateral Takeaways from Japan Pharma Bus Tour

During the MS Japan Pharma bus tour we met with 9 companies (including Chugai, Otsuka, Shionogi, Takeda) over two days in Tokyo. We recap key takeaways as well as laterals for our coverage including ALMS, BMY, GILD, JNJ, LLY, REGN, and VRTX. For US SMID Cap Biotech, multiple laterals also observed.

Otsuka takeaways (VRTX lateral)

Voyxact (APRIL inhibitor): The IgAN launch is tracking ahead of expectations, and 1Q26 sales imply about ~1k doses, or more if you include the bridge program. While they acknowledged some uncertainties on the trajectory given steps between start forms and coverage, they expect sales to grow strongly and broadly don't see any differences from other immunological drugs regarding uptake.

They expressed confidence in **Voyxact interim Ph3 GFR data** the upcoming ERA conference but unclear what time cut will be presented. However, we will likely need to wait for the 2 year data (top-line June-August, followed by a journal publication) to see a cut by ADA status.

They guided to filing an sBLA for full approval based on eGFR data in 2H26, as well as filing for auto-injector approval. Beyond IgAN they are currently focused on a Ph2 trial for Sjogren's syndrome.

Regarding the global IgAN market they pointed to China as an interesting commercial opportunity beyond the US, noting typically there are ~2-3x times more patients in China than US/EU in general.

See takeaways note from our Japan team HERE. *See within for our additional takeaways.*

MORGAN STANLEY & CO. LLC

Terence C Flynn, Ph.D.
Equity Analyst
Terence.Flynn@morganstanley.com +1 212 761-2230

Sean Laaman, Ph.D.
Equity Analyst
Sean.Laaman@morganstanley.com +1 212 761-4947

MORGAN STANLEY MUFG SECURITIES CO., LTD.+

Shinichiro Muraoka
Equity Analyst
Shinichiro.Muraoka@morganstanleymufg.com +81 3 6836-5424

Jaeheon Lee
Equity Analyst
Jaeheon.Lee@morganstanleymufg.com +81 3 6836-8443

Biopharma Takeaways

Santen takeaways (REGN lateral)

Eylea HD conversion in Japan stands at just under 20%, and Santen has guided to 50% of Eylea sales by F3/30.

With respect to potential **M&A/business development**, eye disease is a focus including back of eye, front of eye and glaucoma. Size would be below %0bn yen and IRR >20%. MFN considerations come up more frequently now in these discussions. Long-acting VEGF approaches are on their radar, but a drug/MOA that represents a full replacement to Eylea would be more interesting.

Shionogi takeaways (GILD lateral)

Long-acting HIV treatment program: Reiterated timelines for Cabo Q4M launch at the beginning of 2028 and VH184 (long-acting integrase) for treatment around 2030. They noted there are a number of considerations when evaluating profile of a long-acting regimen including the number of injections, the injection volume, SQ vs. IM injection, and overall injection side effects. They expressed confidence in the profile of ‘184 as they think it will change the treatment market.

Regarding potential market dynamics for long-acting treatment options they noted that they believe Q2M/Q4M/Q6M regimens can ultimately capture ~15-20%/30%/50% of the treatment market.

Long-acting HIV prevention (PrEP): They noted that injectables are taking shares from orals, and that GILD's Yeztugo (Q6M) is not having an impact on Apretude (Q2M) at this point. They believe that a Q4M formulation of Apretude should launch sooner than 2028. They are not optimistic about the PrEP commercial opportunity beyond the US.

Takeda takeaways (ALMS, JNJ and LLY lateral)

Zaso (oral TYK2i): Ph3 psoriasis data were better than they expected and they see efficacy as in range with other oral drugs such as ALMS's Envu (TYK2i) and JNJ's Icotyde (IL23i). Mgmt noted that QD dosing and lack of food effect as advantages over oral competition, but how much this impacts prescribing behavior and patient compliance still TBD. No visibility on label yet, but would argue for differentiated label relative to BMY's Sotyktu (oral TYK2i) and they haven't seen any worrying trends here (i.e., triglycerides and TB).

They expect the oral PsO market to expand – noted current advanced mix is 85% IV and 15% orals, see this moving to 30-40% orals, representing a 3x of the number of patients on oral therapy. Peak sales guidance of $3-$6bn assume orals reach ~33% and Zaso captures a "meaningful" piece of that.

Zaso long-term safety data from ‘3003 trial expected over near-term, which is gating to

NDA filing. Takeda plans to issue a PR when they file for approval rather than for '3003 data. Data from a head-to-head trial vs. Sotyktu are also expected over the near-term, but are not gating to filing, rather more important as a commercial tool. Inclusion of these data in the filing package will depend on discussions with FDA.

Zaso data from two Ph2 IBD trials are expected within the fiscal year, and hence could come in 1Q27 rather than 4Q26. They plan to wait for data from both trials before releasing top-line data as a decision about the Ph3 program will be made holistically. They believe Zaso should work mechanistically as they have better selectivity than first-gen TYK2's if you can dose high enough, but they have not seen any Ph2 data yet so not a slam dunk. The Ph2 IBD trials are exploring higher undisclosed doses than the psoriasis dose (30mg) and they believe they have lots of headroom here (Nimbus previously explored a top dose of 100mg).

Oveporexton (oral orexin receptor 2 agonist): Expressed confidence in the drug's profile, noting it delivers a step change in efficacy and QOL metrics relative to oxybates. They plan to focus the initial launch as monotherapy rather than add on to oxybates. They projected that there are ~100k NT1 patients in the US that could potentially be on therapy, of which 15-17k are already on oxybates, so the question is not only how many patients will switch off of oxybates (more than 80% have residual symptoms), but also how do you increase the overall treatment rate to 40-50%. They are also interested in expanding the use of orexin agonists into other indications (NT2/IH, sleep apnea, cognition and metabolic disorders). In terms of differentiation vs. competitor programs, they noted in their view it will be hard to do better than Oveporexton in NT1, but Takeda will not have an NT2 indication for the drug so maybe competitor drugs might have success here and hence a broader label. But under this scenario dosing/pricing considerations could come into play, hence why Takeda is pursuing development of a portfolio of orexin agonists.

Japan Pharma and US SMID Cap Biotech - linkages

Exhibit 1: Overlap Map of Japanese Pharmaceutical companies on our bus tour and our US SMID Cap Biotech Coverage

JP company	MS-covered overlap (tickers/names called out)	Program / mechanism overlap (as written)	Relationship	Why US investors care (as written)
Takeda	CNTA; AXSM; SLN; plus "JAZZ / ALKS-adjacent; ALMS / NRIX / I&I names"	OX2R agonists (**oveporexton / TAK-861**) vs CNTA orexin franchise; AXSM acquired Takeda-origin CNS asset (**balipodect / TAK-063**); PV read-across (**rusfertide**) vs SLN **divesiran**; immunology read-across (**zasocitinib / TAK-279**)	Direct competitor + partnered/divested asset	"Densest overlap" and commercial benchmark for orexin class; clear read-through to AXSM/Takeda-origin asset; PV class comparator for SLN; oral immunology positioning relevance.
Chugai	VKTX; GPCR; AARD; GUTS; "obesity names"; ASND (muscle); "ALNY/hemophilia-adjacent less direct"	**Orforglipron** (Chugai-origin oral GLP-1 licensed to Lilly); **GYM329** anti-latent myostatin / muscle preservation; **NXT007** next-gen Hemlibra-like bispecific	Partnered / competitive thematic	"Best Japan-originated global innovation read-through": oral GLP-1 economics informs the obesity tape; muscle preservation impacts obesity combo debate; NXT007 informs durability/dosing/switch dynamics in heme.
Otsuka	CGEM; CMPS; ZBIO; AXSM	**Zipalertinib** (Taiho/Cullinan) EGFR exon20 NSCLC; **sibeprenlimab** (APRIL) in IgAN; "Transcend" psychedelics/psychiatry acquisition; "INQOVI + venetoclax AML"	Direct partnership + thematic	"Real direct MS-covered overlap" via Cullinan/Taiho zipalertinib; IgAN/renal-autoimmune relevance; psychiatry capital allocation relevance to CNS coverage .
Nippon Shinyaku	SRPT; DYN; RGNX; RCKT; RARE; PTCT	DMD exon-skipping franchise: **viltepso / viltolarsen**, "NS-051/NCNP-04 exon 51", "NS-089/NCNP-02 exon 44"	Direct competitor / class read-across	"Direct DMD overlap": ability to defend exon-skipping amid gene-therapy volatility, biomarker skepticism, and evolving FDA evidence bars is a major read-through for MS-covered DMD/gene/RNA names.
Santen	FDMT; APLS; BLTE	Retina asset **RC28-E / eflimrufusp alfa** (VEGF/FGF bispecific) vs anti-VEGF durability / wet AMD gene therapy debates	Competitive thematic	"Cleanest overlap" called out is FDMT 4D-150 durability narrative; informs the broader durability/injection-burden and regional pricing debate in retina.
Shionogi	VIR; AVIR; "GILD-adjacent HIV; INSM anti-infective read-through"	AMR antibiotic **Fetroja/Fetcroja**; COVID antiviral **ensitrelvir**; "ViiV HIV economics"; **Radicava** ALS	Mostly thematic / commercial	"Direct overlap thinner" but infectious disease investability, stewardship/government dynamics, and HIV economics can influence how investors value ID platforms and adjacent exposures.
Ono	ONC; GMAB; BCYC; CGEM; oncology SMID M&A universe	Deciphera assets (Qinlock / Romvimza) and kinase/platform framing; PD-1 royalty exposure (noted as platform/M&A read-across)	M&A / oncology platform read-across	"Mostly oncology-platform / M&A read-across rather than direct MS-covered SMID competition"; relevant as buyer/partner template for US oncology SMID universe.
Sumitomo Pharmaceutical	"Menin class: KURA/SNDX not in MS SMID deck"; INCY; TCRX; SLN; ONC	**Enzomenib / DSP-5336** menin inhibitor in AML; **nuvisertib / TP-3654** PIM1 in MF	Competitive thematic, limited direct MS coverage	Menin overlap exists but "pure-play US comps not in this deck"; MS-covered angle is broader heme/AML/MF adjacency (INCY/TCRX/SLN/ONC)

Source: Company data, Morgan Stanley Research.

The Japan–U.S. biotechnology relationship is fundamentally anchored by cross-border business development, where U.S. SMID cap biotech serves as a primary engine of innovation and Japanese pharma acts as a scaled commercialization and capital provider. Facing a material patent cliff and the need for "pipeline recharge," large pharma—including Japan majors—has increasingly externalized R&D through licensing, co-development, and equity-linked partnerships. This has elevated U.S. SMID cap biotech into a structurally advantaged position as a supplier of differentiated clinical assets and enabling platforms, with deal structures (e.g., shared economics, milestone-heavy agreements) balancing risk and access to innovation. Importantly, this dynamic is no longer bilateral; it sits within a broader global sourcing framework where China-origin assets are increasingly competitive in licensing markets—particularly in oncology and cardiometabolic—forcing both U.S. and Japanese players to broaden sourcing strategies and accelerating the pace and competitiveness of cross-border deal flow.

A second defining linkage is modality specialization and complementary innovation strengths. Japanese pharma has built globally competitive positions in select high-complexity modalities—most notably antibody-drug conjugates (ADCs) and oncology development—while U.S. biotech, particularly within SMID cap, continues to lead in next-generation platform innovation spanning bispecific antibodies, T-cell engagers, and emerging gene and cell-based approaches. This creates a structurally interdependent ecosystem: U.S. SMID biotech generates high-value, early-stage or mid-stage innovation, while Japanese partners contribute development discipline, manufacturing scale—particularly in technically demanding biologics—and global commercialization expertise. Notably, manufacturing and supply capabilities in complex modalities have become strategic differentiators, influencing both partnering dynamics and the ultimate scalability of innovation. Parallel developments in China—where oncology and adjacent modalities are seeing rapid asset generation and licensing activity—further intensify competition, reinforcing the need for Japan–U.S. partnerships to move earlier and execute more efficiently to secure best-in-class assets.

The third core connection is execution speed and the globalization of development, where regulatory alignment and operational efficiency determine how quickly innovation translates into commercial opportunity. The U.S. remains the anchor for late-stage development and commercialization, but long-standing FDA–PMDA collaboration frameworks have reduced friction in multi-regional trials, supporting more unified global development strategies and faster time-to-market. For U.S. SMID biotech, this creates a clear pathway: generate proof-of-concept rapidly, often in partnership, and then leverage global partners—including Japanese pharma—to scale development and commercialization. At the same time, rising competition from China—characterized by faster clinical execution and large, efficient patient enrollment—has introduced a new variable into global development timelines and bargaining power in licensing discussions. The net result is an increasingly competitive, triangular system in which U.S. SMID biotech provides innovation depth, Japan provides disciplined development and global scaling capabilities, and China pressures timelines and economics—collectively reshaping how and where innovation is sourced, developed, and ultimately commercialized.

We provide more detail on select areas of overlap below.

1. Polycythemia Vera - Rusfertide (Takeda), Sapablursen (Ono Pharmaceuticals), Diversiram (Silence Therapeutics) & DISC-3405 (Disc Medicine)

The modern development landscape for polycythemia vera (PV) increasingly reflects a deeply interconnected ecosystem spanning the US, Japan, and China, with each region contributing distinct capabilities across discovery, clinical development, and commercialization. In the US, innovative early-stage science and clinical execution remain central, as illustrated by assets such as rusfertide (developed by Protagonist and partnered with Takeda) and DISC-3405, which is being advanced through US-based clinical trials by Disc Medicine as a novel anti-TMPRSS6 antibody for PV. The RNAi approach is also emerging within this ecosystem through assets such as diversiran, which targets TMPRSS6 via gene silencing, further reinforcing the mechanistic validation of iron regulation as a therapeutic axis in PV and highlighting the growing modality diversity within US-led discovery efforts. Japan plays a critical role as a global commercialization and development hub, exemplified by Ono Pharmaceutical's licensing of sapablursen from Ionis, where Ionis retains responsibility for Phase 2 development while Ono assumes global late-stage development and commercialization, supported by a $280M upfront payment and substantial milestone and royalty economics. China, in turn, is increasingly embedded upstream in the innovation chain, as demonstrated by DISC-3405 (originating from Mabwell), where rights were out-licensed to Disc Medicine for ex-China development, with parallel clinical activities ongoing in China and economic participation retained through royalties and milestones. Taken together, PV drug development highlights a triangular model in which US biotech drives discovery and early clinical work, Japanese pharma scales and globalizes assets, and Chinese biotech contributes emerging innovation and capital-efficient drug sourcing.

- PV is becoming a globalized development model — US innovation, Japanese commercialization scale, and China-originated assets are increasingly integrated within single product lifecycles.
- **Rusfertide** defines the near-term benchmark, but durability, dosing burden, and real-world adherence will determine whether it fully captures the PV market.
- **Sapablursen** represents Japan's strategic pivot to RNA platforms, with Ono effectively arbitraging US discovery into global hematology leadership via capital deployment.
- **DISC-3405** highlights China's rising role in upstream innovation, with assets licensed ex-China while retaining regional economics and parallel development tracks.
- **SLN124** was internally generated in the US by Silence Therapeutics using their siRNA platform.
- Key battleground = mechanism + convenience + durability — hepcidin biology is validated, but differentiation will come from dosing interval, depth of hematocrit control, and chronic tolerability.

Comparison of PV programs - snapshot

Exhibit 2: Comparison of PV programs - Takeda (Rusfertide), Ono Pharmaceuticals (ONO-0530), Disc Medicine (DISC-3405) & Silence Therapeutics (SLN124)

Program	Modality / MoA	Target biology	Dosing (studied)	Development stage in PV	"Proof" signal to date	Key safety flags to watch	Commercial timing (evidence-based)
Rusfertide	**Hepcidin mimetic peptide**	Directly increases functional hepcidin → iron restriction → ↓ erythrocytosis	**Weekly SC**, add-on to SOC	**Phase 3 completed; NDA accepted w/ Priority Review**	**VERIFY**: higher response (absence of phlebotomy eligibility) and fewer phlebotomies vs placebo; Hct control and PROs improved	**Anemia** + injection-site reactions (mostly low grade); class needs ongoing thrombosis/MDS/AML vigilance	**PDUFA goal date Q3 2026** (per NDA acceptance/priority review communication)
Sapablursen (ONO-0530)	**ASO to TMPRSS6** (reduces TMPRSS6 → ↑ endogenous hepcidin)	Hepcidin induction via **TMPRSS6 knockdown**	**Q4W SC**	**Phase 2 positive; Phase 3 starting 2026**	**IMPRSSION (Ph2a)**: weekly phlebotomy rate decreased; Hct lowered; hepcidin increased; symptom score improved in higher-dose cohort	Low ISR rate; 1 AML transformation death judged unrelated; monitor iron-restriction anemia	**Phase 3 start ~Jun 2026; primary completion ~Sep 2028; study completion ~2031.**
DISC-3405	**mAb to TMPRSS6** (blocks TMPRSS6 → ↑ endogenous hepcidin)	Hepcidin induction via **TMPRSS6 inhibition**	**SC**; Phase 2 is dose-escalation	**Phase 2 ongoing (open-label)**	**Phase 1 HV**: dose-related ↑ hepcidin, ↓ serum iron (deep, sustained), and reductions in Hb/Hct; Phase 2 PV readout awaited	Expect iron-restriction anemia risk; early HV tolerability favorable	**Phase 2 PV started Aug 2025; primary completion est. Feb 2027; company has guided to initial PV data by end-2026.**
SLN124	**siRNA to TMPRSS6 (silences TMPRSS6 → ↑ endogenous hepcidin)**	Hepcidin induction via TMPRSS6 gene silencing (liver)	Phase 1: Q6W SC; Phase 2: Q6W & Q12W (6 mg)	**Phase 2 ongoing (randomized, double-blind, placebo-controlled)**	Phase 1 SANRECO update: phlebotomies "essentially eliminated"; mean Hct maintained ≤45%; ↑ hepcidin & ferritin;	well tolerated/no DLT Platelet increases (monitor magnitude/clinical sequelae); iron-restriction anemia;	**Phase 2 SANRECO fully enrolled (48 PD-PV); topline anticipated Q3 2026 / "expected August 2026" (company update)**

Source: Company Data, Morgan Stanley Research.

Management commentary (program-by-program)

Rusfertide - Takeda emphasize unmet need, the centrality of controlling Hct <45%, and the ability to reduce phlebotomy burden while improving symptoms; topline messaging highlights no new safety findings and predominance of mild injection-site reactions. NDA acceptance/priority review communications reinforce the regulatory momentum and position rusfertide as potentially first-in-class.

Sapablursen - Ono explicitly call out the physical/psychological burden of phlebotomy and state they "look forward to advancing…in a Phase 3 study," supported by IMPRSSION's reductions in phlebotomy rate and Hct control.

DISC-3405 - Disc communications describe ongoing Phase 2 PV and guide to data from PV and SCD studies expected in Q4 2026.

2. Takeda/Jazz Pharmaceuticals: Orexin Agonists vs Oxybates: Paradigm Shift in Narcolepsy

Mechanism Shift: From Symptom Management to Disease Restoration: Takeda positions orexin agonists—led by oveporexton—as the first therapies to address the root cause of narcolepsy (orexin deficiency), contrasting with oxybates and stimulants that manage symptoms. Management highlights Phase 3 results showing broad improvements in excessive daytime sleepiness (EDS), cataplexy, and quality of life, with many patients approaching normal function. This has driven a clear narrative: orexin agonists may reset standard of care by moving treatment toward mechanism-based therapy.

Jazz Positioning: Co-Existence, Not Replacement: Jazz management's strategy centers on preserving the role of oxybates by emphasizing their unique efficacy in disrupted nighttime sleep (DNS). The company argues that in a post-orexin world, patients may require combination therapy—orexins for daytime symptoms and oxybates for nighttime control. However, this framing is viewed by at least some KOLs as increasingly challenged. Competing companies and some KOLs argue oxybates induce non-physiologic sedation, whereas orexin agonists may normalize overall sleep-wake biology, potentially reducing reliance on nightly sedatives.

Takeda Strategy: From Setback to Leadership: Takeda's program evolution reinforces the credibility of the class. While TAK-994 demonstrated strong efficacy but failed due to hepatotoxicity, oveporexton has shown a clean Phase 3 safety profile, supporting the view that the mechanism is viable. Management emphasized this transition as validation of orexin biology and platform potential, with additional pipeline assets targeting NT2 and idiopathic hypersomnia (IH) to expand beyond NT1.

Efficacy Differentiation: Step-Change vs Oxybates: Clinical data increasingly support Takeda's narrative: Oxybates: ~15–18 min improvement in MWT vs Orexin agonists: ~25–35 min improvement (near ceiling). This magnitude of benefit underpins the view that orexins may represent a step-change in efficacy in NT1.

Current Oxybate Reality: Effective but Constrained. Despite strong efficacy, oxybates face structural limitations: i) Complex dosing (overnight administration); ii) Adherence challenges and side effects; & iii) Sodium and neuropsychiatric burden. These factors contribute to low penetration (~25% treated), highlighting a market that is both underpenetrated and vulnerable to disruption.

Indication-Specific Impact: NT1 vs NT2 vs IH. i) NT1 (highest disruption): Orexin deficiency aligns with mechanism; orexins likely to become first-line, with potential to replace oxybates in many patients. ii) NT2 (moderate impact): Intact orexin biology; efficacy still meaningful but less differentiated → gradual share shift; iii) IH (expansion story): Less direct competition; orexins may expand diagnosis and treatment, increasing overall market size.

Orexin agonists—led by Takeda—represent the first credible mechanism-based disruption of the oxybate franchise, with the greatest risk in NT1. While Jazz's co-existence strategy may hold in the near term, the weight of data and biology increasingly supports a shift toward orexin-first treatment, with oxybates transitioning from a cornerstone therapy to a secondary or adjunct role within a significantly expanded market.

3. Otsuka/Taiho & Cullinan Therapeutics (CGEM) on Zipalertinib

Japan–US biotech linkage and broader partnering implications. Zipalertinib is a textbook example of bi-directional innovation flow between Japan and the U.S. SMID biotech ecosystem. The molecule was discovered within Taiho in Japan, leveraging its internal chemistry and discovery platforms, while clinical development, capital formation, and investor engagement have been heavily U.S.-driven through Cullinan. This mirrors a broader trend we've highlighted: Japanese pharma increasingly externalizes development and commercialization partnerships to access U.S. capital markets, regulatory expertise, and oncology trial infrastructure, while U.S. SMID biotechs leverage Japan-originated assets to build pipelines without fully bearing discovery risk. The opt-in co-promotion structure for Cullinan further illustrates how these partnerships are evolving from simple licensing deals into more integrated, shared-risk commercialization models, especially as assets approach the U.S. market.

Recent commentary across Cullinan Therapeutics (CGEM), Taiho Pharmaceutical / Taiho Oncology, and your internal channel checks highlights a program moving into a clearly defined regulatory and commercial inflection window, with a differentiated positioning in EGFR exon20 NSCLC that is increasingly being framed around tolerability, oral convenience, and line-of-therapy flexibility.

Near-term events and regulatory momentum. The program has transitioned from data validation to regulatory execution. Following positive Phase 2b REZILIENT1 data presented at ASCO 2025, Taiho and Cullinan initiated and subsequently completed a rolling NDA submission to the FDA, with the application formally accepted and a PDUFA date of February 27, 2027. This cadence aligns with prior management commentary (NDA filing in 2H25, frontline Phase 3 REZILIENT3 enrollment completing ~1H26) and positions approval and launch sequencing into a 2026–2027 window. Additional clinical catalysts (REZILIENT2 expansion cohorts, frontline Phase 3 data) reinforce a multi-year data build into earlier lines, a dynamic management explicitly tied to combination strategies with chemotherapy and potential best-in-class positioning.

CGEM/Taiho view on differentiation and market opportunity. Across investor events and KOL commentary, the consistent framing is that zipalertinib's opportunity is less about absolute efficacy delta and more about usability and tolerability within the treatment algorithm. In REZILIENT1, zipalertinib demonstrated ~35% ORR with durable responses and a manageable safety profile in heavily pre-treated patients. Cullinan management and KOLs have emphasized comparability of efficacy even in post-amivantamab patients (~30% ORR) alongside a more favorable tolerability and oral route, which is expected to drive meaningful physician and patient preference versus IV-based options. This supports a commercial narrative where zipalertinib could be used sequentially after chemo ± amivantamab, but importantly may also capture patients opting out of infusion-based regimens, creating incremental demand rather than purely share shift. From a sizing standpoint, our CGEM modeling points to ~$390M unadjusted peak sales (1L + 2L combined) pre-Taiho economics, with contribution split across second-line (~$160M) and first-line (~$235M) settings. While modest in the context of broader NSCLC, management commentary consistently highlights a high unmet need niche (EGFR exon20ins) with limited well-tolerated targeted options, reinforcing a durable, specialty-market profile rather than a commoditized TKI dynamic.

Investor questioning has not yet fully converged on zipalertinib - suggesting the market may be underappreciating the asset's near-term relevance relative to broader Otsuka priorities. This creates a potential setup where regulatory milestones (NDA acceptance, PDUFA visibility) could drive increased focus and re-rating as commercialization strategy becomes clearer.

4. Nippon Shinyaku, Sarepta, Biomarin & RegenxBio in DMD

Nippon Shinyaku's DMD strategy—spanning its in-house exon-skipping franchise (e.g., Viltepso) and its partnership with Capricor on CAP-1002—serves as a clear bridge between the Japanese and U.S. biotechnology ecosystems by combining Japan's strength in rare disease commercialization and regulatory access with U.S.-led innovation in advanced modalities like cell therapy and gene-based approaches. Through its U.S. subsidiary NS Pharma and exclusive commercialization agreements with Capricor, Nippon Shinyaku effectively imports U.S. clinical innovation into both the U.S. and Japanese markets, while also contributing global development and commercialization infrastructure tailored to rare neuromuscular diseases. This model highlights a broader structural dynamic we've focused on across our coverage—where Japanese pharma increasingly acts as a capital-efficient development and commercialization partner for U.S. biotech assets, particularly in areas like DMD where patient populations are small, global coordination is critical, and regulatory alignment across regions can accelerate uptake.

Exhibit 3: DMD programs - Nippon Shinyaku, Sarepta, RegenxBio & Biomarin

Company	Asset	Modality / MOA	Stage / Status	Key Biomarker Data	Clinical Outcomes	Safety	Positioning
Sarepta	Elevidys (SRP-9001)	AAV gene therapy delivering micro-dystrophin	Approved (US; ambulatory), Ph3 confirmatory	Micro-dystrophin expression used as surrogate endpoint	Functional benefit signals (NSAA improvements vs control; mixed randomized data)	Liver toxicity risk; serious AEs incl myocarditis	First-mover gene therapy; standard-setter
Regenxbio	RGX-202	Next-gen AAV gene therapy (optimized micro-dystrophin)	Pivotal; topline data expected 2026	Biomarker + functional + cardiac MRI endpoints reported in early trials	Interim functional benefit signals; readout pending	TBD; positioning on improved safety vs SRPT	Potential best-in-class GT challenger
BioMarin	BMN-351	Exon-skipping oligonucleotide (next-gen)	Early clinical (Ph1/2)	Dystrophin expression at ~25 weeks (biopsy endpoint)	POC data pending (functional endpoints TBD)	Class-related liver/kidney signals being monitored	Competes within exon-skipping niche
Nippon Shinyaku / Capricor	CAP-1002 (deramiocel)	Cell therapy (cardiosphere-derived cells; immunomodulatory/exosome-mediated)	Ph3 complete; PDUFA Aug 2026	Not dystrophin-based; targets inflammation/fibrosis	Ph3: significant improvement in upper limb function + cardiac (LVEF)	Favorable tolerability	Adjunct / mutation-agnostic therapy

Source: Company data, Morgan Stanley Research

Across Duchenne muscular dystrophy (DMD), the four programs we highlight map into two fundamentally different "disease-modifying" strategies—dystrophin restoration (gene therapy or exon skipping) versus mutation-agnostic functional preservation (cell/exosome biology)—and that split largely determines both their addressable populations and commercial ceilings. In major markets, prevalence is often cited at roughly ~31K patients across the 7MM with ~17K in the U.S., underscoring that even small shifts in eligibility and penetration can drive meaningful revenue outcomes in a rare-disease setting.

Sarepta is the current commercial anchor via Elevidys (SRP-9001), an AAV micro-dystrophin gene therapy that uses micro-dystrophin expression as a surrogate endpoint* to support approval logic. While the clinical narrative remains debated—earlier reporting highlights mixed randomized functional outcomes (e.g., NSAA) alongside supportive biomarker and external-control arguments—Sarepta's scale advantage is already evident in the P&L: ~$898.7M of Elevidys net product revenue in 2025 (with additional ~PMO exon-skipping franchise revenue reported separately). Sarepta also continues to present accumulating long-term efficacy and safety experience across its DMD portfolio at major congresses, reflecting the growing importance of durability and real-world safety characterization for sustained penetration.

Regenxbio's RGX-202 is best framed as the most direct "next-gen" gene therapy challenger to Sarepta's position. The company has been signaling pivotal-level progress and has hosted investor/physician-facing events specifically tied to topline results from a pivotal trial of RGX-202 (announced May 2026). In practice, the competitive bar for RGX-202 is straightforward: to earn meaningful share against an entrenched first mover, it likely needs to demonstrate either a superior benefit/risk profile, clearer functional translation, and/or operational advantages in the real-world treatment pathway—though the specific topline dataset itself is not contained in the announcement source.

BioMarin's BMN-351 sits in a different lane: next-generation exon skipping. From our KOL discussions, BMN-351 is described as having completed enrollment across dose cohorts, with planned presentation of initial proof-of-concept data including ~25-week muscle biopsy dystrophin data from early treated patients. Exon skipping's structural limitation is that it tends to be mutation-/exon-specific, which can cap the eligible pool versus mutation-agnostic gene therapy; consequently, BMN-351's commercial upside is more likely tied to a defined genetic subset and to differentiation on dystrophin expression magnitude, dosing convenience, and tolerability within the exon-skipping class (the specific comparative performance versus competitors is not provided in the cited source).

Nippon Shinyaku's most investable "DMD program" in this comparison is effectively CAP-1002 (deramiocel) via Capricor/NS Pharma, which is mutation-agnostic and does not depend on dystrophin restoration. Instead, it is positioned around immunomodulatory/anti-fibrotic biology (exosome-mediated mechanisms) aimed at preserving skeletal and cardiac muscle function. Critically, this program has reported a pivotal Phase 3 (HOPE-3) as meeting its primary endpoint (upper limb function, PUL v2.0) and a key cardiac endpoint (LVEF), and the regulatory path is tangible: the FDA resumed review and set a PDUFA date of Aug 22, 2026. Strategically, that profile supports CAP-1002 as a potential add-on "backbone" for later-stage disease or cardiomyopathy-focused benefit, rather than a direct substitute for dystrophin-restoring approaches.

From a market lens, third-party market research sources peg the global DMD treatment market at ~$4.8bn (recently cited) with continued growth driven by gene therapy and molecular approaches, broadly consistent with multi-year expansion narratives. The near- to mid-term competitive dynamic is therefore less about whether DMD is a large market (it is, for rare disease) and more about how the market stratifies: (1) one-time gene therapy concentrating early in eligible pediatric populations with safety/durability as gating variables (Sarepta now; Regenxbio attempting to follow), (2) mutation-defined chronic exon skipping as a narrower but potentially durable niche where next-gen candidates must prove biomarker and functional differentiation (BioMarin's BMN-351), and (3) mutation-agnostic, function/cardiac-centric adjunct therapy that can layer onto dystrophin strategies and address later-stage unmet need (CAP-1002/NS Pharma).

Disclosure Section

The information and opinions in Morgan Stanley Research were prepared by Morgan Stanley & Co. LLC, and/or Morgan Stanley C.T.V.M. S.A., and/or Morgan Stanley Mexico, Casa de Bolsa, S.A. de C.V., and/or Morgan Stanley Canada Limited. As used in this disclosure section, "Morgan Stanley" includes Morgan Stanley & Co. LLC, Morgan Stanley C.T.V.M. S.A., Morgan Stanley Mexico, Casa de Bolsa, S.A. de C.V., Morgan Stanley Canada Limited and their affiliates as necessary.

For important disclosures, stock price charts and equity rating histories regarding companies that are the subject of this report, please see the Morgan Stanley Research Disclosure Website at www.morganstanley.com/eqr/disclosures/webapp/generalresearch, or contact your investment representative or Morgan Stanley Research at 1585 Broadway, (Attention: Research Management), New York, NY, 10036 USA.

For valuation methodology and risks associated with any recommendation, rating or price target referenced in this research report, please contact the Client Support Team as follows: US/Canada +1 800 303-2495; Hong Kong +852 2848-5999; Latin America +1 718 754-5444 (U.S.); London +44 (0)20-7425-8169; Singapore +65 6834-6860; Sydney +61 (0)2-9770-1505; Tokyo +81 (0)3-6836-9000. Alternatively you may contact your investment representative or Morgan Stanley Research at 1585 Broadway, (Attention: Research Management), New York, NY 10036 USA.

Analyst Certification

The following analysts hereby certify that their views about the companies and their securities discussed in this report are accurately expressed and that they have not received and will not receive direct or indirect compensation in exchange for expressing specific recommendations or views in this report: Terence C Flynn, Ph.D.; Sean Laaman, Ph.D.; Jaeheon Lee; Shinichiro Muraoka.

.

Global Research Conflict Management Policy

Morgan Stanley Research has been published in accordance with our conflict management policy, which is available at www.morganstanley.com/institutional/research/conflictpolicies. A Portuguese version of the policy can be found at www.morganstanley.com.br

Important Regulatory Disclosures on Subject Companies

The equity research analysts or strategists principally responsible for the preparation of Morgan Stanley Research have received compensation based upon various factors, including quality of research, investor client feedback, stock picking, competitive factors, firm revenues and overall investment banking revenues. Equity Research analysts' or strategists' compensation is not linked to investment banking or capital markets transactions performed by Morgan Stanley or the profitability or revenues of particular trading desks.

Morgan Stanley and its affiliates do business that relates to companies/instruments covered in Morgan Stanley Research, including market making, providing liquidity, fund management, commercial banking, extension of credit, investment services and investment banking. Morgan Stanley sells to and buys from customers the securities/instruments of companies covered in Morgan Stanley Research on a principal basis. Morgan Stanley may have a position in the debt of the Company or instruments discussed in this report. Morgan Stanley trades or may trade as principal in the debt securities (or in related derivatives) that are the subject of the debt research report.

Certain disclosures listed above are also for compliance with applicable regulations in non-US jurisdictions.

STOCK RATINGS

Morgan Stanley uses a relative rating system using terms such as Overweight, Equal-weight, Not-Rated or Underweight (see definitions below). Morgan Stanley does not assign ratings of Buy, Hold or Sell to the stocks we cover. Overweight, Equal-weight, Not-Rated and Underweight are not the equivalent of buy, hold and sell. Investors should carefully read the definitions of all ratings used in Morgan Stanley Research. In addition, since Morgan Stanley Research contains more complete information concerning the analyst's views, investors should carefully read Morgan Stanley Research, in its entirety, and not infer the contents from the rating alone. In any case, ratings (or research) should not be used or relied upon as investment advice. An investor's decision to buy or sell a stock should depend on individual circumstances (such as the investor's existing holdings) and other considerations.

Global Stock Ratings Distribution

(as of April 30, 2026)

The Stock Ratings described below apply to Morgan Stanley's Fundamental Equity Research and do not apply to Debt Research produced by the Firm.

For disclosure purposes only (in accordance with FINRA requirements), we include the category headings of Buy, Hold, and Sell alongside our ratings of Overweight, Equal-weight, Not-Rated and Underweight. Morgan Stanley does not assign ratings of Buy, Hold or Sell to the stocks we cover. Overweight, Equal-weight, Not-Rated and Underweight are not the equivalent of buy, hold, and sell but represent recommended relative weightings (see definitions below). To satisfy regulatory requirements, we correspond Overweight, our most positive stock rating, with a buy recommendation; we correspond Equal-weight and Not-Rated to hold and Underweight to sell recommendations, respectively.

	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
Stock Rating Category	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1546	42%	467	51%	30%	709	44%
Equal-weight/Hold	1568	43%	358	39%	23%	715	44%
Not-Rated/Hold	4	0%	0	0%	0%	1	0%
Underweight/Sell	555	15%	84	9%	15%	202	12%
Total	3,673		909			1627	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

Analyst Stock Ratings

Overweight (O). The stock's total return is expected to exceed the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Equal-weight (E). The stock's total return is expected to be in line with the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Not-Rated (NR). Currently the analyst does not have adequate conviction about the stock's total return relative to the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Underweight (U). The stock's total return is expected to be below the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Unless otherwise specified, the time frame for price targets included in Morgan Stanley Research is 12 to 18 months.

Analyst Industry Views

Attractive (A): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be attractive vs. the relevant broad market benchmark, as indicated below.

In-Line (I): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be in line with the relevant broad market benchmark, as indicated below.

Cautious (C): The analyst views the performance of his or her industry coverage universe over the next 12-18 months with caution vs. the relevant broad market benchmark, as indicated below.

Benchmarks for each region are as follows: North America - S&P 500; Latin America - relevant MSCI country index or MSCI Latin America Index; Europe - MSCI Europe; Japan - TOPIX; Asia - relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Important Disclosures for Morgan Stanley Smith Barney LLC Customers

Important disclosures regarding any material conflict of interest that can reasonably be expected to have influenced Morgan Stanley Smith Barney LLC's choice of a third-party research provider or the subject company of a third-party research report, are available on the Morgan Stanley Wealth Management disclosure website at www.morganstanley.com/online/researchdisclosures. For Morgan Stanley specific disclosures, you may refer to https://www.morganstanley.com/eqr/disclosures/webapp/generalresearch.

Each Morgan Stanley research report is reviewed and approved on behalf of Morgan Stanley Smith Barney LLC. This review and approval is conducted by the same person who reviews the research report on behalf of Morgan Stanley. This could create a conflict of interest.

Other Important Disclosures

Morgan Stanley Research policy is to update research reports as and when the Research Analyst and Research Management deem appropriate, based on developments with the issuer, the sector, or the market that may have a material impact on the research views or opinions stated therein. In addition, certain Research publications are intended to be updated on a regular periodic basis (weekly/monthly/quarterly/annual) and will ordinarily be updated with that frequency, unless the Research Analyst and Research Management determine that a different publication schedule is appropriate based on current conditions.

Morgan Stanley is not acting as a municipal advisor and the opinions or views contained herein are not intended to be, and do not constitute, advice within the meaning of Section 975 of the Dodd-Frank Wall Street Reform and Consumer Protection Act.

Morgan Stanley produces an equity research product called a "Tactical Idea." Views contained in a "Tactical Idea" on a particular stock may be contrary to the recommendations or views expressed in research on the same stock. This may be the result of differing time horizons, methodologies, market events, or other factors. For all research available on a particular stock, please contact your sales representative or go to Matrix at http://www.morganstanley.com/matrix.

Morgan Stanley Research is provided to our clients through our proprietary research portal on Matrix and also distributed electronically by Morgan Stanley to clients. Certain, but not all, Morgan Stanley Research products are also made available to clients through third-party vendors or redistributed to clients through alternate electronic means as a convenience. For access to all available Morgan Stanley Research, please contact your sales representative or go to Matrix at http://www.morganstanley.com/matrix.

Any access and/or use of Morgan Stanley Research is subject to Morgan Stanley's Terms of Use (http://www.morganstanley.com/terms.html). By accessing and/or using Morgan Stanley Research, you are indicating that you have read and agree to be bound by our Terms of Use (http://www.morganstanley.com/terms.html). In addition you consent to Morgan Stanley processing your personal data and using cookies in accordance with our Privacy Policy and our Global Cookies Policy (http://www.morganstanley.com/privacy_pledge.html), including for the purposes of setting your preferences and to collect readership data so that we can deliver better and more personalized service and products to you. To find out more information about how Morgan Stanley processes personal data, how we use cookies and how to reject cookies see our Privacy Policy and our Global Cookies Policy (http://www.morganstanley.com/privacy_pledge.html). Please use the provided link to review the Terms and Conditions and Most Important Terms and Conditions for Morgan Stanley India Company Private Limited (https://www.morganstanley.com/assets/pdfs/about-us-global-offices/india/Terms_and_conditions.pdf) and the following link to review the audit report (https://ny.matrix.ms.com/eqr/research/webapp/researchdocs/MSICPL_Morgan_Stanley_Research_Audit_Report.pdf).

If you do not agree to our Terms of Use and/or if you do not wish to provide your consent to Morgan Stanley processing your personal data or using cookies please do not access our research.

Morgan Stanley Research does not provide individually tailored investment advice. Morgan Stanley Research has been prepared without regard to the circumstances and objectives of those who receive it. Morgan Stanley recommends that investors independently evaluate particular investments and strategies, and encourages investors to seek the advice of a financial adviser. The appropriateness of an investment or strategy will depend on an investor's circumstances and objectives. The securities, instruments, or strategies discussed in Morgan Stanley Research may not be suitable for all investors, and certain investors may not be eligible to purchase or participate in some or all of them. Morgan Stanley Research is not an offer to buy or sell or the solicitation of an offer to buy or sell any security/instrument or to participate in any particular trading strategy. The value of and income from your investments may vary because of changes in interest rates, foreign exchange rates, default rates, prepayment rates, securities/instruments prices, market indexes, operational or financial conditions of companies or other factors. There may be time limitations on the exercise of options or other rights in securities/instruments transactions. Past performance is not necessarily a guide to future performance. Estimates of future performance are based on assumptions that may not be realized. If provided, and unless otherwise stated, the closing price on the cover page is that of the primary exchange for the subject company's securities/instruments.

The fixed income research analysts, strategists or economists principally responsible for the preparation of Morgan Stanley Research have received compensation based upon various factors, including quality, accuracy and value of research, firm profitability or revenues (which include fixed income trading and capital markets profitability or revenues), client feedback and competitive factors. Fixed Income Research analysts', strategists' or economists' compensation is not linked to investment banking or capital markets transactions performed by Morgan Stanley or the profitability or revenues of particular trading desks.

The "Important Regulatory Disclosures on Subject Companies" section in Morgan Stanley Research lists all companies mentioned where Morgan Stanley owns 1% or more of a class of common equity securities of the companies. For all other companies mentioned in Morgan Stanley Research, Morgan Stanley may have an investment of less than 1% in securities/instruments or derivatives of securities/instruments of companies and may trade them in ways different from those discussed in Morgan Stanley Research. Employees of Morgan Stanley not involved in the preparation of Morgan Stanley Research may have investments in securities/instruments or derivatives of securities/instruments of companies mentioned and may trade them in ways different from those discussed in Morgan Stanley Research. Derivatives may be issued by Morgan Stanley or associated persons.